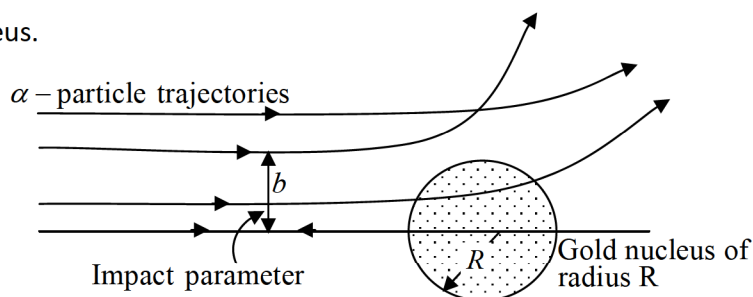


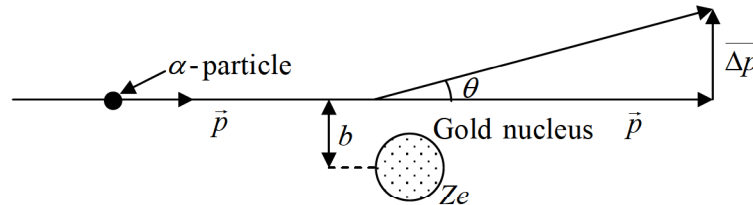
Chapter One

Basic Nuclear Properties

1.1 Rutherford Experiment and Properties of Nucleus

You are all familiar with the structure of an atom. The modern picture of an atom tells us that there is a core called the nucleus, which contains all the positive charge, and almost all the mass of the atom. This nucleus has positively charged particles called the protons in it, and electrically neutral neutrons. In the first part of this course it is this core called the nucleus that will occupy our time. In an atom, which is electrically neutral there is equal amount of negative charge in it. The negatively charged electrons are all distributed in different atomic orbitals around this nucleus.





Principle of Rutherford experiment:

By bombarding a very thin gold foil with α -particles, there were an observation that a small fraction of these α -particles were deflected at large angle.

This angle is defined b/w the asymptotic direction of incident particle and asymptotic line of deported α -particle which is known as scattering angle (θ).

And the minimum distance at which α -particle would approach the nucleus if there were no force between them is known as impact parameter (b)

And on this impact parameter and radius of the nucleus there is dependence of change in deflection, which is categorized as:-

- For $b < r$ (Ray 2), the deflection produced in the α -particle trajectory is small.
- For $b > r$ (Ray 4) due to less coulomb repulsion the deflection of α -particle is also less.
- And for $b \sim r$ (Ray 3) the deflection of α -particle becomes maximum.
- There is also a deflection at one point (Ray 1) where the KE of α -particle will start converting into the coulombian potential energy and at one point its KE will become 0 and that point will come back due to repulsive force and the distance b/w that point and nuclei is known as distance of closest approach.

Conclusion:

- Most of the space inside the atom is empty.
- As a few α -particles got deviated through small angles there must be some positively charged particles in matter which repel the α -particles strongly causing deviation and there positively charged particles known as nucleus.
- As a very small number of α -particles got deviated through large angles and retraced their path, the size of the nucleus must be very small compared to that of an "atom".
The nucleus must also be very heavy.

Calculation of the repulsive force

$$F = \frac{e^2}{4\pi\epsilon_0} \cdot \frac{2z}{b^2}, \quad \frac{e^2}{4\pi\epsilon_0} = 1.44 \text{ MeV-fm}$$

Calculation of distance of closest approach (x)

$$K = \frac{e^2}{4\pi\epsilon_0} \cdot \frac{2z}{x}$$

Calculation of impact parameter:

$$b = \frac{ze^2}{4\pi\epsilon_0 k} \cot \frac{\theta}{2}$$

where k is the KE of α -particle and θ is scattering angle.

Angular scattering distribution.

$$N(\theta) = \frac{N_i n L z^2 k^2 e^4}{4r^2 KE^2 \sin^4 \frac{\theta}{2}}$$

where N_i = number of incident α -particles

n = atoms per unit volume in target

L = thickness of target

z = atomic number of target

e = electronic charge

k = coulomb's constant

r = target to deflector distance

KE = KE of α -particle

θ = scattering angle.

Experiments on scattering of α particle on gold foil shows there is existence of nucleus at the centre of atom. The experiments demonstrate radius of a nucleus is smaller than the radius of atom by factor of about 10^4 . This means the volume of an atom is about 10–12 times the volume of the nucleus. In other words, an atom is almost empty.

Atomic mass: The mass of atom is very small compare to kilogram .for example mass of carbon atom ^{12}C is $1.99 \times 10^{-26} \text{ kg}$.Kilogram is not a very convenient unit to measure such small quantities .therefore a different mass unit is used for expressing atomic masses. This unit is the atomic mass unit (u), defined as $\frac{1}{12}$ of the mass of the carbon ^{12}C

$$1u = \frac{\text{mass of one } ^{12}\text{C atom}}{12} = \frac{1.99 \times 10^{-26}}{12} = 1.66 \times 10^{-27} \text{ kg}$$

Example: Mass of iron is $55.85u$. What is equivalent mass in kg?

Solution: Mass of iron is $55.85u = 55.85 \times 1.66 \times 10^{-27} \text{ kg} = 9.26 \times 10^{-26} \text{ kg}$

Composition of atom and nucleus: classical model of atoms suggest **any** atom consist of electron and nucleus. Electron is negatively charged particle with mass $m_e = 0.00055u$. According to classical theory electron is orbiting about nucleus under influence of electrostatic force. The nucleus consists of Proton and neutron. Proton is discovered by Earnest Rutherford. Proton is positively charged same magnitude of electron and it has mass of $m_p = 1.0072u = 1.67 \times 10^{-27} \text{ kg}$ which is 1836-time mass of an electron. The hydrogen atom has only proton as its nucleus. Although over 99.9 percent of hydrogen nuclei are just single protons, a few also contain another particle named as neutron, and a very few two neutrons, along with the protons. The neutron is discovered by Chadwick. Neutron is neutral particle and it has mass $m_n = 1.0086u = 1.6749 \times 10^{-27} \text{ kg}$ which is almost same as mass of proton.

Other than hydrogen atom all other atoms has proton and neutron in its nucleus.

The conventional symbols for nuclear species, or nuclides, follow the pattern ${}_Z^AX$, where X = chemical symbol of the element.

Z = atomic number of the element which is equal to number of protons in the nucleus

A = mass number of the nuclide number of nucleons in the nucleus.

That means mass number is equal to number of protons plus number of neutrons.

Number of neutron $N = A - Z$

Mass Defect: Mass defect is the difference between the actual atomic mass and the predicted mass calculated by adding the mass of protons and neutrons present in the nucleus. The actual atomic mass is less than the predicted mass calculated by adding the masses of nucleons. This additional mass is accounted for by binding energy that is released when a nucleus is formed. When a nucleus is formed, some of the mass is converted to energy and this result in the mass defect. Due to this reason, the actual mass of an atomic nucleus is less than the mass of particles it is made up of.

The mass defect is given by $\Delta M = (Zm_p + Nm_n) - M_A$ where Z is number of protons N is number of neutrons and M_A actual mass of nucleus.

Mass Energy Equivalence: Einstein showed from his theory of special relativity that it is necessary to treat mass as another form of energy Einstein gave the famous mass-energy

equivalence relation $E = mc^2$ where m is relativistic mass of particle and c is speed of light which is given by $3 \times 10^8 \text{ m/sec}$. If rest mass is given by m_0 . The relativistic mass is given by

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

the total energy is given by $E = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$

Relativistic momentum is given by Where $p = mv = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$

In term of momentum p the total relativistic energy is given by $E = \sqrt{p^2 c^2 + m_0^2 c^4}$.

1gm of substance is equivalent to $1 \times 10^{-3} \text{ kg} \times (3 \times 10^8)^2 = 9 \times 10^{13} \text{ J}$ which means if 1g of matter is converted into energy, there is release of $9 \times 10^{13} \text{ J}$ of energy.

1 Joule is equivalent to $6.24 \times 10^{18} \text{ eV} = 6.24 \times 10^{12} \text{ MeV}$. So 1g of mass is equivalent to

$$9 \times 10^{13} \text{ J} \times 6.24 \times 10^{18} = 56 \times 10^{31} \text{ eV}$$

1 atomic mass unit (1u) is equivalent to $\frac{931 \text{ MeV}}{c^2}$

Example: A Gold nucleus has chemical composition ${}^{197}_{79}\text{Au}$.

- Find the atomic number, mass number, number of proton and number of neutron.
- Find the predicted mass of ${}^{197}_{79}\text{Au}$ nucleus in atomic mass unit. Find the predicted mass of ${}^{197}_{79}\text{Au}$ in kg.
- If actual mass of ${}^{197}_{79}\text{Au}$ is $196.9665u$, find value of mass defect in kg.
- Find the value of mass defect in unit of $\frac{\text{MeV}}{c^2}$.
- Find the value of mass defect in MeV.

Solution: (a) atomic number $Z = 79$

Mass number $A = 197$

Number of protons = atomic number $Z = 79$

Number of neutrons $N = A - Z = 197 - 79 = 118$

(b) The mass of nucleus is

$$79 \times m_p + 118 \times m_n = 79 \times 1.00727u + 118 \times 1.00866u = 198.5962u$$

Mass of nucleus of ${}^{197}_{79}\text{Au}$ in kg $198 \times 1.67 \times 10^{-27} = 331.6 \times 10^{-27} \text{ kg} = 33.16 \times 10^{-26} \text{ kg}$

(c) Predicted mass $198.5962u$ actual mass is $196.9665u$

Mass defect is $\Delta M = 198.596u - 196.966u = 1.624u$

$$= 1.62 \times 1.67 \times 10^{-27} \text{ kg} = 2.62 \times 10^{-27} \text{ kg} = 0.2 \times 10^{-26} \text{ kg}$$

(d) If mass defect is $1.624u$ which is equivalent to $\frac{1.62 \times 931.5}{c^2} = \frac{1509 \text{ MeV}}{c^2}$

(e) equivalent energy for mass defect $1.624u \times c^2 = \frac{1509 \text{ MeV}}{c^2} \times c^2 = 1509 \text{ MeV}$

Some rest masses in various units are:

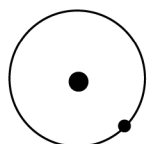
Particle	Mass(kg)	Mass(u)	Mass(MeV/c ²)
Proton	1.6726×10^{-27}	1.007276	938.28
Neutron	1.6750×10^{-27}	1.008665	939.57
Electron	9.1095×10^{-31}	5.486×10^{-4}	0.511
${}_1\text{H}^1$	1.6736×10^{-27}	1.007825	938.79

Nuclear Terminology

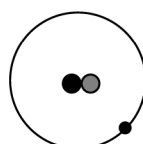
- Isotopes:** If two nuclei have same atomic number Z (proton), then they are called as isotopes. Example: ${}^{13}_6\text{C}$ & ${}^{14}_6\text{C}$, ${}^{16}_8\text{O}$ & ${}^{17}_8\text{O}$ and ${}^1_1\text{H}$, ${}^2_1\text{H}$, ${}^3_1\text{H}$

The varieties of an element that differ in the numbers of neutrons their nuclei contain are called **isotopes**.

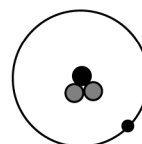
Ordinary hydrogen



Deuterium



Tritium



Proton



Neutron



Electron

Figure: The isotope of hydrogen

- Isotones:** If two nuclei have same neutron number N (proton), then they are called as isotones. Example: ${}^{13}_6\text{C}$ and ${}^{14}_7\text{N}$
- Isobars:** If two nuclei have same mass number A , then they are called as isobars. Example: ${}^{14}_6\text{C}$ and ${}^{14}_7\text{N}$
- Mirror nuclei:** Nuclei with same mass number A but with proton and neutron number interchanged and their difference is ± 1 .

Example: ${}^{11}_6\text{C}$ & ${}^{11}_5\text{B}$ and ${}^{13}_7\text{N}$ & ${}^{13}_6\text{C}$

Size of the nucleus

Radius of Nucleus: By performing scattering experiments in which fast electrons, instead of α -particles, are projectiles that bombard targets made up of various elements, the sizes of nuclei of various elements have been accurately measured. It has been found that a nucleus of mass number A has a radius $R = R_0 A^{1/3}$ where $R_0 = 1.2 \times 10^{-15} \text{ m} = 1.2 \text{ fermi}$.

Volume of Nucleus: This means the volume of the nucleus, which is proportional to R^3 is proportional to A .

Mass Density of Nucleus: Thus the density of nucleus is a constant, independent of A , for all nuclei. Different nuclei are like drop of liquid of constant density.

Assuming spherical symmetry, volume of nucleus is given by $V = \frac{4}{3} \pi R^3 = \frac{4}{3} \pi R_0^3 A$.

Mass of one proton = $1.67 \times 10^{-27} \text{ kg}$, Nuclear Mass = $A \times 1.67 \times 10^{-27} \text{ kg}$.

$$\text{Nuclear density} = \frac{A \times 1.67 \times 10^{-27}}{\frac{4}{3} \pi R_0^3 \times A} = 2.3 \times 10^{17} \text{ kg/m}^3$$

Nucleon Particle Density

Nuclear particle density is ratio of nuclear mass density to nuclear mass per nucleon.

Nuclear mass density is approximately 10^{17} kg/m^3 and nuclear mass per nucleons is $1.67 \times 10^{-27} \text{ kg}$ so

$$\begin{aligned} \text{Nuclear Particle Density} &= \frac{\text{Nuclear Mass Density}}{\text{Nuclear Mass/nucleon}} \\ &= \frac{10^{17} \text{ Kg/m}^3}{1.67 \times 10^{-27} \text{ Kg/Nucleon}} \approx 10^{44} \text{ Nucleons/m}^3 \end{aligned}$$

Example: Given the mass of iron nucleus ${}^{56}_{26}\text{Fe}$ is $55.85u$

- Find radius of nucleus.
- Find volume of nucleus.
- Find mass density of nucleus.
- Find nucleons particle density.

Solution:

(a) Radius is given by $R = R_0 \times A^{1/3}$ where $R_0 = 1.2 \times 10^{-15} \text{ m}$ and $A = 56$

$$1.2 \times 10^{-15} \times (56)^{1/3} = 1.2 \times 10^{-15} \times 3.82 = 4.59 \times 10^{-15} \text{ m}$$

(b) volume of nucleus is $\frac{4\pi}{3} R^3 = \frac{4\pi}{3} \times (R_0)^3 (A^{1/3})^3$

$$= \frac{4\pi}{3} R_0^3 A = \frac{4 \times 3.14}{3} \times (1.2 \times 10^{-15})^3 \times 56 = 405 \times 10^{-45} m^3$$

(c) density is given by $\rho = \frac{m}{V}$

Mass of iron nucleus is $55.85u$ where $u = 1.66 \times 10^{-27} kg$ the mass of iron nucleus is

$$55.8 \times 1.66 \times 10^{-27} kg = 9.27 \times 10^{-26} kg$$

Volume is given by $405 \times 10^{-45} m^3$

Density is $\rho = \frac{m}{V} = \frac{9.27 \times 10^{-26}}{405 \times 10^{-45}} = 0.0228 \times 10^{19} = 2.29 \times 10^{17} kg / m^3$

(d) $n = \frac{\text{Nuclear Mass Density}}{\text{Nuclear Mass/nucleon}} = \frac{2.29 \times 10^{17} kg / m^3}{1.67 \times 10^{-27} kg / nucleon} = 1.37 \times 10^{44} \text{ Nucleon} / m^3$

Example: Electrons of kinetic energy $K = 500 MeV$ are scattered from a target of nuclei of charge distribution radius R into a diffraction pattern that has minima of average separation of $\theta \approx 30^\circ$ estimate the value of R .

Solution: Rest mass of electron is $m_0 c^2 = 0.511 MeV$ which is negligible compare to kinetic energy $K = 500 MeV$

So momentum $pc = E = K \Rightarrow p = \frac{K}{c} = \frac{500 MeV}{3 \times 10^8 m / sec} \times \frac{1 Joule}{6.2 \times 10^{12} MeV} = 2.7 \times 10^{-19} kg - m / sec$

The de Broglie relation gives $\lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34} Joule - sec}{2.7 \times 10^{-19} kg - m / sec} = 2.4 \times 10^{-15} m$

$$R = \frac{\lambda}{\theta} = \frac{2.4 \times 10^{-15}}{0.53 rad} = 4.5 \times 10^{-15} m = 4.5 \text{ Fermi}$$

Uncertainty principle and radius of nucleus.

Using Heisenberg uncertainty principle, it can be shown why electrons can not be within nucleus.

$\Delta x \Delta p_x \approx \hbar \Rightarrow \Delta p_x = \frac{\hbar}{\Delta x}$ the uncertainty in position can be equal to radius of nucleus so

$$\Delta x \approx R = 10^{-14} \Delta p_x = \frac{6.6 \times 10^{-34}}{10^{-14}} = 6.6 \times 10^{-20} \Rightarrow p_x = 6.6 \times 10^{-20} kg - m / sec$$

$$E = \sqrt{p_x^2 c^2 + m_0^2 c^4} \Rightarrow E = \sqrt{(6.6 \times 10^{-20} \times 3 \times 10^8 \times 6.24 \times 10^{12})^2 + (0.511)^2} \text{ MeV}$$

$\sqrt{(124)^2 + (0.511)^2} = 124 \text{ MeV}$ which is very large energy to achieve this energy electron need to have speed more than light so it cannot exist within nucleus

1.2 Fermi Model

A multi fermion system at zero kelvin temperature

Fermions cannot occupy the same state, even at zero temperature. At zero temperature the first fermion sets in the lowest state in the potential well, each next fermion sets in the lowest free state in the well.

Consequently, n fermions occupy the n lowest-energy states in the potential well. The energy of the highest occupied state is called the Fermi energy.

At zero temperature all states up to the Fermi energy are occupied, while all states above the Fermi energy are free. The total energy of the system is equal to the sum of energies of the occupied states.

Assumption of Fermi gas model

The Fermi gas model defines properties of a system of non-interacting fermions in an infinite potential well.

The model predicts gross properties of various quantum-mechanical systems, for example electrons in metals, or nucleons in nuclei.

The model assumes that all fermions occupy the lowest energy states available to them up to the Fermi energy, and that there is no excitations across the Fermi energy (i.e. zero temperature).

In nuclei the model assumes that protons and neutrons are independent fermion filling two separate potential wells. The model assumes, however, common Fermi energy for the protons and neutrons in stable nuclei.

If Fermi energy for protons and neutrons are different then the β decay transforms one type of nucleons into the other until the common Fermi energy (stability) is reached.

Mathematical discussion of model

If a particle of mass m is confined in infinite box with parameter (L_x, L_y, L_z) then energy is

given by $E_{n_x, n_y, n_z} = \frac{\pi^2 \hbar^2}{2m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right)$ where $n_x = 1, 2, 3 \dots$ $n_y = 1, 2, 3 \dots$ $n_z = 1, 2, 3 \dots$

for simplicity the $L_x = L_y = L_z = L$ $E_{n_x, n_y, n_z} = \frac{\pi^2 \hbar^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2)$,

where the momentum is given $p_x = \frac{n_x \pi \hbar}{L}$ $p_y = \frac{n_y \pi \hbar}{L}$ $p_z = \frac{n_z \pi \hbar}{L}$

let us consider E_F is fermi energy $E_F = \frac{p_F^2}{2m} \Rightarrow p_F = \sqrt{2mE_F}$

$$p_x^2 + p_y^2 + p_z^2 \leq p_F^2 \Rightarrow \frac{\pi^2 \hbar^2}{L^2} (n_x^2 + n_y^2 + n_z^2) \leq p_F^2 \Rightarrow (n_x^2 + n_y^2 + n_z^2) \leq \frac{p_F^2 L^2}{\pi^2 \hbar^2} = R^2$$

where $R = \frac{p_F L}{\pi \hbar}$ assume to be radius of sphere if (n_x, n_y, n_z) are representing coordinates along

x, y and z The volume of the sphere is the number of states up to the Fermi

momentum since there is one state per unit volume.

To count number of state (n_x, n_y, n_z) we use a trick First, remember that $n_x > 0$ and $n_y > 0$ and $n_z > 0$, thus we have to take only $1/8$ of the sphere cut by positive x, y and z axes.

Next, nucleons have spins. Thus we have two states per unit volume as defined by two orientation of spins.

With these corrections we can calculate the number of occupied states

$$n = 2 \frac{1}{8} \frac{4}{3} \pi R^3 = \frac{1}{3} \pi \left(\frac{p_F L}{\pi \hbar} \right)^3$$

Calculation of Fermi Momentum

$$p_F = \hbar (3\pi^2)^{1/3} \frac{n^{1/3}}{L} = \hbar (3\pi^2)^{1/3} \left(\frac{n}{L^3} \right)^{1/3} = \hbar (3\pi^2)^{1/3} \left(\frac{n}{V} \right)^{1/3}$$

$$\text{If } R = R_0 A^{1/3} \quad V = \frac{4}{3} \pi (R_0 A^{1/3})^3 = \frac{4}{3} \pi R_0^3 A \quad \text{if there is } Z \text{ number of protons}$$

$$\text{Fermi momentum for protons } p_F = \hbar (3\pi^2)^{1/3} \left(\frac{n}{V} \right)^{1/3} = \frac{\hbar}{R_0} \left(\frac{3^2 \pi}{4} \right)^{1/3} \left(\frac{Z}{A} \right)^{1/3}$$

$$\text{Fermi momentum for neutrons } p_F = \hbar (3\pi^2)^{1/3} \left(\frac{n}{V} \right)^{1/3} = \frac{\hbar}{R_0} \left(\frac{3^2 \pi}{4} \right)^{1/3} \left(\frac{A-Z}{A} \right)^{1/3}$$

Where Z is atomic number and A is mass number

Assuming $Z/A \sim 1/2$ and $R_0 = 1.2 \text{ fm}$ the estimate for the Fermi momentum is

$$p_F = \frac{\hbar}{R_0} \left(\frac{3^2 \pi}{4} \right)^{1/3} \left(\frac{Z}{A} \right)^{1/3} \approx 250 [MeV/c] \text{ and speed } \beta_f = \frac{v_f}{c} \approx \frac{p_F c}{m_p c^2} \approx \frac{250}{940} = 0.27$$

Estimation of Fermi Energy

Fermi energy can be estimated from the Fermi momentum.

Fermi energy depends on the density of states and is different for protons and for neutrons since in general there is a different number of particles in the proton and neutron well.

The Fermi energy for protons is $E_F = \frac{p_F^2}{2m_p} = \frac{(\hbar c)^2}{2R_0^2 m_p c^2} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \left(\frac{Z}{A} \right)^{\frac{2}{3}}$

The Fermi energy for neutron is $E_f = \frac{p_F^2}{2m_p} = \frac{(\hbar c)^2}{2R_0^2 m_p c^2} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \left(\frac{A-Z}{A} \right)^{\frac{2}{3}}$

For the case of even number of protons and neutrons $Z/A \approx (A-Z)/A \approx 1/2$ and $R_0 = 1.2 fm$ the Fermi energy is $E_f \sim 33 MeV$.

Spin and Magnetic Moment

Proton and neutrons, like electrons, are fermions with spin quantum numbers of $s = \frac{1}{2}$. This

means they have spin angular momenta $\vec{S}$ of magnitude $|\vec{S}| = \sqrt{s(s+1)}\hbar = \sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)}\hbar = \frac{\sqrt{3}}{2}\hbar$

and spin magnetic quantum number of $m_s = \pm \frac{1}{2}$.

As in case of electrons, magnetic moments of nucleus are associated with the spins of protons and neutrons. In nuclear physics magnetic moments are expressed in **nuclear magnetons** (μ_N),

where nuclear magneton $\mu_N = \frac{e\hbar}{2m_p} = 5.051 \times 10^{-27} J/T = 3.152 \times 10^{-8} eV/T$ and m_p is the proton mass.

In atomic physics we have defined **Bohr magnetons** $\mu_B = \frac{e\hbar}{2m_e}$ where m_e is the electron mass.

The nuclear magneton is smaller than the Bohr magneton by the ratio of the proton mass to the electron mass which is 1836.

$$\frac{\mu_B}{\mu_N} = \frac{m_p}{m_e} = 1836.$$

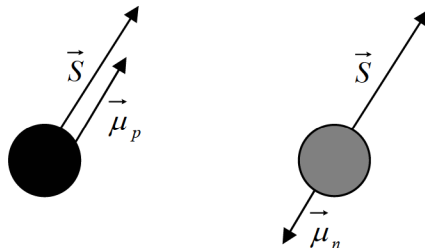
The spin magnetic moments of the proton and neutron have components in any direction of

Proton $\mu_{pz} = +2.793\mu_N$

Neutron $\mu_{nz} = \mp 1.913\mu_N$

There are two possibilities for the signs of μ_{pz} and μ_{nz} , depending on whether m_s is $-\frac{1}{2}$ or $+\frac{1}{2}$.

The $\pm$ sign is used for μ_{pz} because $\vec{\mu}_{pz}$ is in the same direction as the spin $\vec{S}$, whereas $\mp$ is used for μ_{nz} because $\vec{\mu}_{nz}$ is opposite to spin $\vec{S}$.



Note: For neutron, magnetic moment is expected to be zero as $e = 0$, but $\mu_n = \mp 1.913\mu_N$.

At first glance it seems odd that the neutron, with no net charge, has spin magnetic moment. But if we assume that the neutron contains equal amounts of positive and negative charge, a spin magnetic moment arises if these charges are not uniformly distributed. The magnetic moment of neutron further explained by quark model.

Two type of quark are necessary to build proton and neutrons, there are up quark with charge of $\frac{2}{3}e$ and the down quark with $-\frac{1}{3}e$. The proton consists of two up quark and one down quark so that total charge $+e$ and neutron on other hand is built up of one up quark and 2 down quark so that its total charge is zero but there is magnetic dipole moment due to magnetic moment of individual quark.

Thus, we can say that neutron has physical significance of negative charges because magnetic moment is opposite to that of its intrinsic spin angular momentum.

Angular Momentum of Nucleus

The hydrogen nucleus ^1_1H consists of a single proton and its total angular momentum is given by $|\vec{S}| = \frac{\sqrt{3}}{2}\hbar$. A nucleon in a more complex nucleus may have orbital angular momentum due to motion inside the nucleus as well as spin angular momentum. The total angular momentum of such a nucleus is the vector sum of the spin and orbital angular momenta of its nucleons, as in the analogous case of the electrons of an atom.

Chapter Two

Binding Energy

It has been seen that the predicted mass $(Zm_p + (A - Z)m_n)$ is more than the actual mass M .

This additional mass is accounted for by binding energy that is released when a nucleus is formed.

The difference in predicted mass and actual mass is identified as mass defect ΔM .

For any nucleus composition A_ZX mass defect is $\Delta M = [Zm_p + (A - Z)m_n] - M$

If we want to break the nucleus of A_ZX into Z number of proton and $A - Z$ number of neutrons we need to supply ΔMc^2 amount to energy to nucleus. The energy equivalent to mass defect is identify as Binding energy $E_b = \Delta mc^2$

The convenient measure to analyse Binding energy of nucleus ratio between binding energy E_b and its constituents i.e. A . If we plot binding energy per nucleon $\frac{E_b}{A}$ as number of mass number we get following curve.

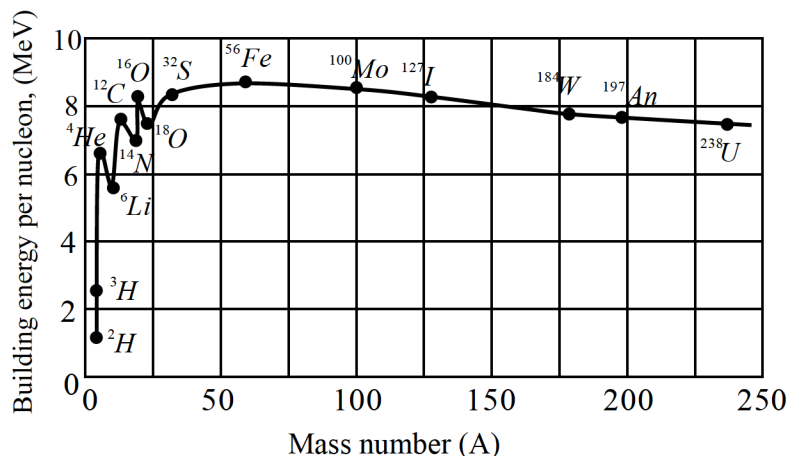


Figure: The binding energy per nucleon as a function of mass number

Main feature of Binding energy per nucleon versus the mass number

(a) The binding energy per nucleon, $\frac{E_b}{A}$, is practically constant, i.e. practically independent of the atomic number for nuclei of middle mass number ($30 < A < 170$). The curve has a maximum of about 8.75 MeV for $A = 56$ MeV and has a value of 7.6 MeV for $A = 238$

(b) $\frac{E_b}{A}$ is lower for both light nuclei $A < 30$ and $A > 170$

We can draw some conclusions from these two observations:

- (i) The force is attractive and sufficiently strong to produce a binding energy of a few MeV per nucleon.
- (ii) The constancy of the binding energy in the range $30 < A < 170$ is a consequence of the fact that the nuclear force is short-ranged.

Consider a particular nucleon inside a sufficiently large nucleus. It will be under the influence of only some of its neighbours, which come within the range of the nuclear force. If any other nucleon is at a distance more than the range of the nuclear force from the particular nucleon it will have no influence on the binding energy of the nucleon under consideration.

(iii) A very heavy nucleus, say $A = 240$, has lower binding energy per nucleon compared to that of a nucleus with $A = 120$. Thus, if a nucleus $A = 240$ breaks into two $A = 120$ nuclei, nucleons get more tightly bound. This implies energy would be released in the process. It has very important implications for energy production through fission,

(iv) Consider two very light nuclei ($A \leq 10$) joining to form a heavier nucleus. The binding energy per nucleon of the fused heavier nuclei is more than the binding energy per nucleon of

the lighter nuclei. This means that the final system is more tightly bound than the initial system.

Again, energy would be released in such a process of fusion

Example: If mass of ${}^4_2\text{He}$ nucleus is 4.002602u find the binding energy of ${}^4_2\text{He}$. Also calculate

Binding energy per nucleon for ${}^4_2\text{He}$.

Solution: $E_b = \Delta Mc^2 = \left([Zm_p - (A - Z)m_n] - M \right) c^2$

$$E_b = \left[(2 \times 1.0078u + (4 - 2) \times 1.00866u) - 4.002602u \right] c^2 = 0.030378uc^2$$

$$E_b = 0.030378uc^2 \times \frac{931\text{MeV}}{c^2} = 28.3\text{MeV}$$

$$\frac{B.E}{A} = \frac{28.3}{4} = 7.07\text{MeV / nucleon}$$

Example: The measured mass of deuteron atom (${}^2_1\text{H}$), Hydrogen atom (${}^1_1\text{H}$), proton and neutron is 2.01649 u, 1.00782 u, 1.00727 u and 1.00866 u. Then find the binding energy of the deuteron nucleus (unit MeV/nucleon).

Solution: B.E = $[Z M_H + N M_N - M({}^2_1\text{H})] \times 931.5$

$$= [1 \times 1.00782 + 1 \times 1.00866 - 2.01649] \times 931.5$$

$$= (0.00238) \times 931.5 = 2.224 \text{ MeV}$$

Example: The binding energy of the neon isotope ${}^{20}_{10}\text{Ne}$ is 160.647 MeV. Find its atomic mass.

Solution: Here $Z=10$ and $A=10$, $M({}^A_Z\text{X}) = [ZM_H + NM_N] - \frac{B}{931.5 \text{ MeV / u}}$

$$M({}^{20}_{10}\text{Ne}) = \left[10(1.00782) + 10(1.00866) \right] - \frac{160.647}{931.5 \text{ MeV / u}} = 19.992 \text{ u}$$

Example:

(a) Find the energy needed to remove a neutron from the nucleus of the calcium isotope ${}^{42}_{20}\text{Ca}$.

(b) Find the energy needed to remove a proton from this nucleus.

(c) Why are these energies different?

Given: atomic masses of ${}^{42}_{20}\text{Ca} = 41.958622u$, ${}^{41}_{20}\text{Ca} = 40.962278u$, ${}^{41}_{19}\text{K} = 40.961825u$,

and mass of ${}^1_0n = 1.008665u$, ${}^1_1p = 1.007276u$.

Solution:

(a) ${}_{20}^{42}\text{Ca} \rightarrow {}_{20}^{41}\text{Ca} + {}_0^1n$; Total mass of the ${}_{20}^{41}\text{Ca}$ and ${}_0^1n = 41.970943 \text{ u}$.

Mass defect $\Delta m = 41.970943 - 41.958622 = 0.012321 \text{ u}$

So, B.E. of missing neutron $= \Delta m \times 931.5 = 11.48 \text{ MeV}$.

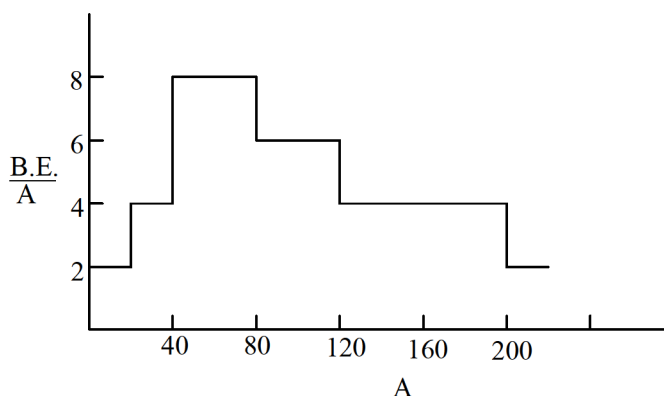
(b) ${}_{20}^{42}\text{Ca} \rightarrow {}_{19}^{41}\text{K} + {}_1^1p$; Total mass of the ${}_{19}^{41}\text{K}$ and ${}_1^1p = 41.919101 \text{ u}$.

Mass defect $\Delta m = 41.919101 - 41.958622 = 0.010479 \text{ u}$

So, B.E. of missing neutron $= \Delta m \times 931.5 = 10.27 \text{ MeV}$.

(c) The neutron was acted upon only by attractive nuclear forces whereas the proton was also acted upon by repulsive electric forces that decrease its binding energy.

Example: The following histogram represents the binding energy per particle ($B.E./A$) in MeV as a function of the mass number A of a nucleus. A nucleus with mass number $A = 180$ fissions into two nuclei of equal masses.



(a) Find the Binding energy for parents' nucleus $A = 180$

(b) Find the Binding energy of product nucleus (sum of binding energy of individual nucleus)

(c) how much energy is released /absorbed during fission

Solution: $A \rightarrow \frac{A}{2} + \frac{A}{2}$ or $180 \rightarrow 90 + 90$

(a) B.E. of nucleus $A = 180 \times 4 = 720 \text{ MeV}$

(b) Binding energy of Product $= 90 \times 6 + 90 \times 6 = 1080 \text{ MeV}$

(c) Since, B.E of the product nucleus is greater than the nucleus A , hence in this process energy is released and that is $= (1080 - 720) \text{ MeV} = 360 \text{ MeV}$.

Nuclear Force:

The binding energy per nucleon is approximately 8MeV which is much larger than binding energy of atom. There are a variety of experiments carried out during 1930 to 1950 which give the following conclusion about nuclear forces.

- (1) The nuclear force is much stronger than the Coulomb force acting between charges or the gravitational forces between masses.
- (2) The nuclear force between two nucleons falls rapidly to zero as their distance is more than a few femtometres. This leads to saturation of forces in a medium or a large-sized nucleus, which is the reason for the constancy of the binding energy per nucleon.

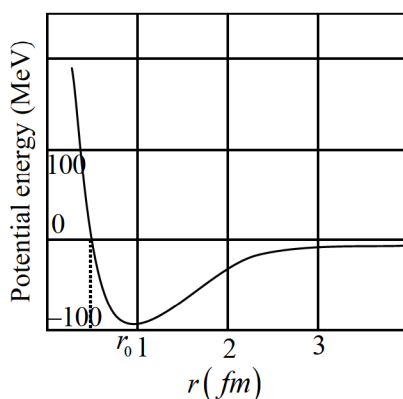


Figure: Potential energy of a pair of nucleons as a function of their separation. For a separation greater than r_0 , the force is attractive and for separations less than r_0 , the force is strongly repulsive.

The best approximation of nuclear potential is Yukawa potential given by Japanese scientist Hideki Yukawa. The Yukawa potential is given by $V(r) = g^2 \frac{\exp(-\alpha mr)}{r}$ where m is mass of

particle g^2 and α is positive constant. For $m = 0$, $g^2 = \frac{q_1 q_2}{4\pi\epsilon_0}$ Yukawa potential will approach to

Coulomb potential

A rough plot of the potential energy between two nucleons

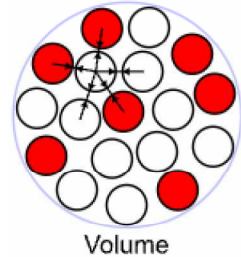
Liquid Drop Model of Binding Energy

This model is given by George Gamow, but finally it is proposed by Carl Friedrich Weizsacker. In theory we will see how nucleus behaves as drop of liquid. The liquid drop accounts for the spherical shape of most of nuclei and makes a rough prediction of binding energy. It is also known as semi-empirical mass formula.

We start assuming that energy associated with each nucleon-nucleon bond has some value U . This energy is negative since attractive forces are involved, but is usually written as positive because binding energy is considered a positive for convenience.

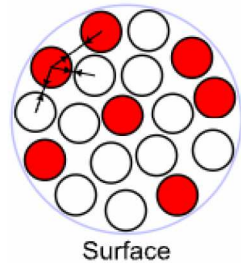
The corresponding mass formula is defined purely in term of number of protons and neutrons it contains. the original Weizsacker formula defines five terms

(1) **Volume Energy:** When an assembly of nucleons of the same size is packed together into the smallest volume, each interior nucleon has a certain number of other nucleons in contact with it. So, this nuclear energy is proportional to the volume. each bound energy is shared by two nucleons and twelve other spherical nucleons are in contact with is so each interior nucleons has binding energy $12 \times \left(\frac{1}{2}\right)U = 6U$.



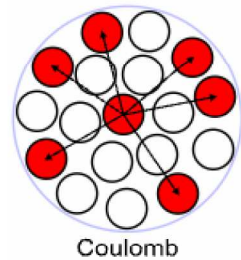
If all nucleons A in a nucleus were in the interior, the total binding energy would be $6AU$ the volume energy of nucleus can be written as $E_V = a_V A$.

(2) **Surface Energy:** Nucleons on the surface of every of every nucleus and therefore have fewer than twelve neighbours. the number of these nucleons depends on surface area of nucleus. The surface area is $4\pi R^2 = 4\pi R_0^2 A^{2/3}$. Hence the number of nucleons with fewer than the maximum number of bonds is proportional to $A^{2/3}$.



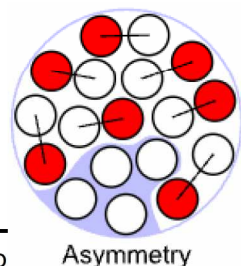
So, surface energy is $E_S = -a_S A^{2/3}$.

(3) **Coulomb Energy:** This term arises due to electrical repulsion between proton and proton. The coulomb energy of nucleus is a work that must be done to bring Z electron from infinity so average coulomb energy is given by $\frac{Z(Z-1)}{2} \cdot \left(-\frac{e^2}{4\pi\epsilon_0} \right) \left\langle \frac{1}{r} \right\rangle$ where $r \propto A^{1/3}$. So coulomb energy is given by



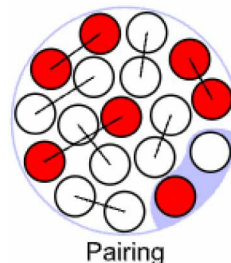
$$-a_C \frac{Z(Z-1)}{A^{1/3}}.$$

(4) **Asymmetry Energy:** (also called Pauli Energy), which accounts for the (Pauli exclusion principle). Unequal numbers of neutrons and protons imply filling higher energy levels for one type of particle, while leaving lower energy levels vacant for the other type.



This is a quantum effect arising from the Pauli exclusion principle which only allows two protons or two neutrons (with opposite spin direction) in each energy state. If a nucleus contains the same number of protons and neutrons then for each type the protons and neutrons fill to the same maximum energy level (the 'fermi level'). If, on the other hand, we exchange one of the neutrons by a proton then that proton would be required by the exclusion principle to occupy a higher energy state, since all the ones below it are already occupied. The upshot of this is that nuclides with $Z = N = A - Z$ have a higher binding energy, whereas for nuclei with different numbers of protons and neutrons (for fixed A) the binding energy decreases as the square of the number difference. The spacing between energy levels is inversely proportional to the volume of the nucleus - this can be seen by treating the nucleus as a three-dimensional potential well- and therefore inversely

Pairing Term: It is found experimentally that two protons or two neutrons bind more strongly than one proton and one neutron. The pairing term is positive for even-even nuclei such as ${}^4_2\text{He}$, ${}^{12}_6\text{C}$, ${}^{16}_8\text{O}$. it is zero for even-odd nuclei and negative for odd-odd nuclei. this term seems to vary as $A^{-3/4}$



So binding energy according to Weizsacker is

$$E_B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} - (\pm, 0) \frac{a_p}{A^{3/4}}$$

The coefficient that give good data is

$$a_v = 14.1 \text{ MeV} \quad a_s = 13.0 \text{ MeV} \quad a_c = 0.595 \text{ MeV} \quad a_A = 19 \text{ MeV} \quad a_p = 33 \text{ MeV}$$

Last two term Asymmetric and pairing term is correction to liquid drop model finally known as Semi-empirical mass formula for Binding energy.

Stability of nucleus

Example: In the liquid drop model the binding term due to coulomb repulsion is $-a_c \frac{Z(Z-1)}{A^{1/3}}$,

find the value of a_c

Solution: This term represent electric repulsion of the proton is negative term in binding energy.

The charge density $\rho = \frac{Ze}{\frac{4\pi}{3} R^3}$

Consider a thin spherical shell of this charge of radius $r < R$ using coulomb's law we can calculate the work dW required to bring this shell up to its radius r

$$dW = \frac{1}{4\pi\epsilon_0 r} \left[\frac{4}{3} \pi r^3 \rho \times 4\pi r^2 dr \rho \right]$$

$$W = \frac{1}{4\pi\epsilon_0} \frac{(4\pi)^2}{3} \left(\frac{Ze}{\frac{4\pi}{3} R^3} \right)^2 \int_0^R r^4 dr = \frac{3}{5} \frac{Z^2 e^2}{4\pi\epsilon_0 R}$$

In the equation of binding energy $B = (Zm_p + Nm_n - M)c^2$ it is assume that there is not already Z number of proton existed in the nucleus .

When we have only one proton in nucleus the $(B_c)_{proton} = \frac{3}{5} \frac{e^2}{4\pi\epsilon_0 R}$

For nucleus only one proton No work is done against the coulomb repulsion in assembling the nucleus.

In calculation of work done $W = \frac{3}{5} \frac{Z^2 e^2}{4\pi\epsilon_0 R}$ those term is also there which is binding energy of

single protons .so we must subtract Z time such a term

So binding energy due to coulomb repulsion is

$$\begin{aligned} -B_c &= \left(\frac{3Z^2 e^2}{5(4\pi\epsilon_0)R} - \frac{3}{5} \frac{Ze^2}{4\pi\epsilon_0 R} \right) = -\frac{3}{5} \frac{e^2 Z(Z-1)}{4\pi\epsilon_0 R} \\ \frac{3}{5} m_e c^2 \frac{e^2}{4\pi\epsilon_0 m_e c^2} \frac{Z(Z-1)}{1.2 \times A^{1/3}} \text{ where } m_e c^2 &= 0.511 \text{ MeV} \\ \frac{e^2}{4\pi\epsilon_0 m_e c^2} &= \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(9.11 \times 10^{-31})(3 \times 10^8)^2} = 2.8 \times 10^{-15} \text{ m} \end{aligned}$$

$$\text{So } -B_c = \frac{3}{5} \times 0.511 \times \frac{2.8 \times 10^{-15}}{1.2 \times 10^{-15}} = 0.66 \text{ MeV}$$

Example: Using formula of Binding energy, find the most stable isobar for $A = 25$.

Solution: $E_b = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} (\pm, 0) \frac{a_p}{A^{3/4}}$

$$\frac{dE_b}{dZ} = 0 \Rightarrow \frac{a_c}{A^{1/3}} (2Z-1) + \frac{4a_A}{A} (A-2Z) = 0$$

$$Z = \frac{a_c A^{-1/3} + 4a_A}{2a_c A^{-1/3} + 8a_A A^{-1}} = \frac{0.6 A^{-1/3} + 76}{1.19 A^{-1/3} + 152 A^{-1}}$$

For $A = 25$, $Z = 11.7 \approx 12$

For $A = 25$ there are three nuclide ${}^{25}_{11}\text{Na}$, ${}^{25}_{12}\text{Mg}$ and ${}^{25}_{13}\text{Al}$

So most stable is ${}^{25}_{12}\text{Mg}$.

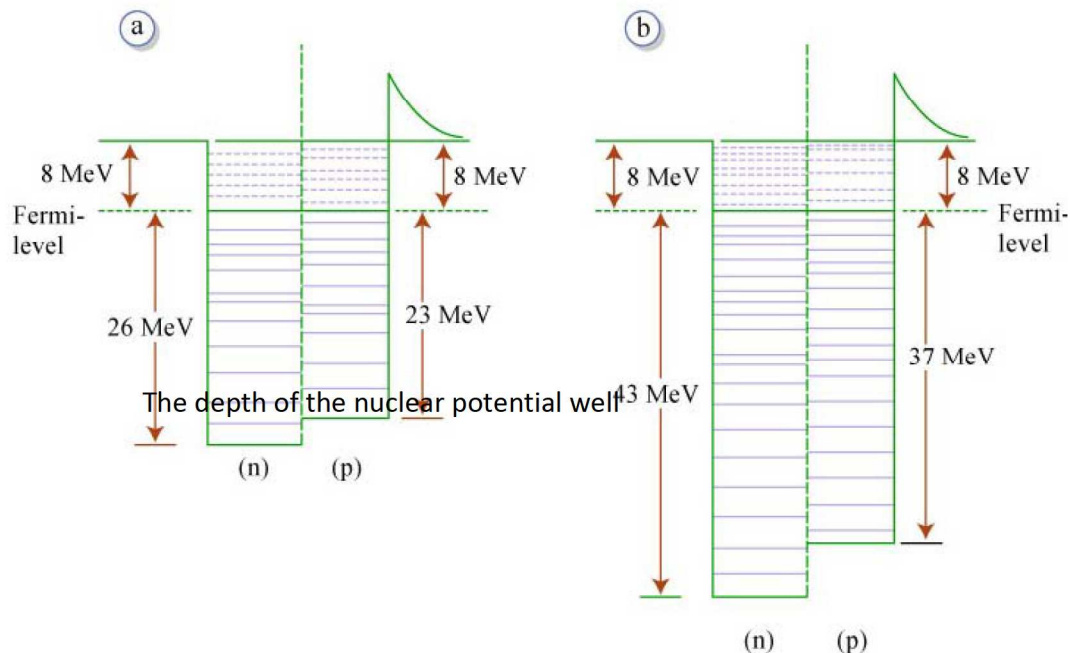
The depth of the nuclear potential well

From the Fermi energy we know (approximately) the energy difference between the bottom of the potential well and the energy of the highest occupied state.

We also know from the binding energies that the last nucleon is bound by approximately 8 MeV

From that we can conclude that the depth of the nuclear potential well is approximately $33 + 8 = 41$

We get the same depth for protons and neutron since we assumed the equal number of both. If we take into account the fact that the difference in the number of protons and neutrons we will get different depths of the potential wells for each species.



The depth difference for protons and neutrons

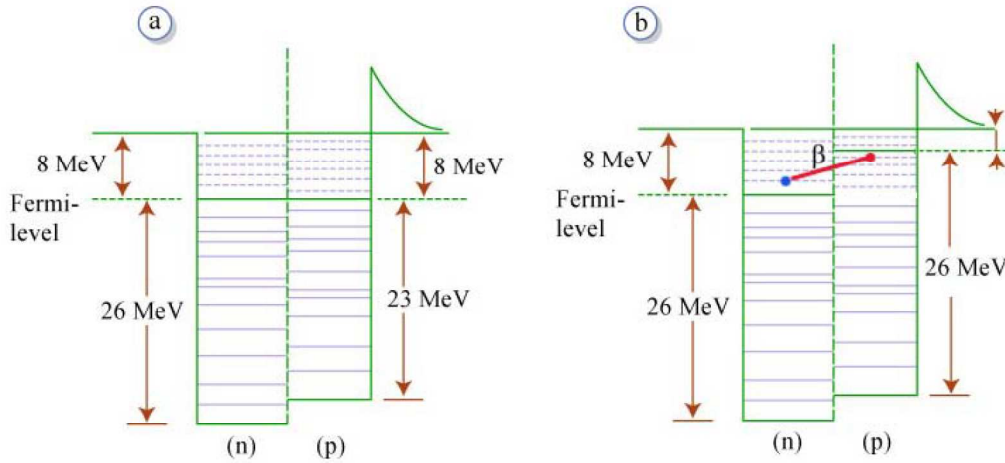
- (1) The potential depth difference comes out from the dependence of the Fermi energy on the number of protons or neutrons.
- (2) The difference in depth is a consequence of the Coulomb repulsion between the protons.

Beta decay and fermi energy

In stable nuclei the Fermi level is at the same energy for protons in neutrons. In such a case there is no energy gain from transforming one type of nucleons into another by β -decay.

In unstable nuclei the Fermi level is different for protons and neutrons, this opens a path to transform nucleons from one well to the other through β -decay. The β decay proceeds until Fermi levels in both wells are equal.

Fermi level for protons in neutrons in stable nuclei (a) and in unstable nuclei (b).



Average momentum of a nucleon

The average momentum of a nucleon depends on the density of momentum states which is the number of momentum states dn for a particle with momentum p and $p + dp$.

We can calculate this number using a similar trick to what we used to calculate the number of states up to the Fermi momentum.

Let us consider the space defined by the quantum numbers (n_x, n_y, n_z) . We argued there are two momentum states per unit volume in this space.

Particles with momentum between p and $p + dp$ define a spherical shell with radius $R = pL / \pi \hbar$ and thickness $dR = dpL / \pi \hbar$

$$\text{in this space with volume } 4\pi R^2 dR = 4\pi \left(\frac{L}{\pi \hbar} \right)^3 p^2 dp = \frac{4V}{\pi^2 \hbar^3} p^2 dp$$

To get the number of states we need to remember that the condition for $n_x > 0, n_y > 0$ and $n_z > 0$ requires that we take $1/8$ of the volume V , and that we also need to multiply V by 2 to account for two spins state of the nucleon.

The density of states is

$$dn = 2 \frac{1}{8} V = \frac{1}{4} \frac{4V}{\pi^2 \hbar^3} p^2 dp = \frac{V}{\pi^2 \hbar^3} p^2 dp \Rightarrow \frac{dn}{dp} = \frac{V}{\pi^2 \hbar^3} p^2$$

The average momentum of a nucleon is

$$p_{av} = \frac{\int_0^{p_F} p \frac{dn}{dp} p^2 dp}{\int_0^{p_F} \frac{dn}{dp} p^2 dp} = \frac{\int_0^{p_F} p^3 dp}{\int_0^{p_F} p^2 dp} = \frac{3}{4} p_F \approx 188 \text{ MeV}$$

The average energy of a nucleon

To calculate the average energy, we need to know the density of energy state, which is the number of states for a particle with energy between E and dE .

We can get this number from the density of the momentum states

$$E = \frac{p^2}{2m} \Rightarrow p^2 = 2mE \Rightarrow p dp = m dE \text{ and } dp = \frac{m dE}{\sqrt{2mE}} = \sqrt{\frac{m}{2}} \frac{dE}{\sqrt{E}}$$

$$dn = \frac{V}{\pi^2 \hbar^3} p^2 dp \quad dn = \sqrt{2} \frac{V}{\pi^2 \hbar^3} m^{3/2} \sqrt{E} dE$$

The average energy of a nucleon

The average energy of a nucleon is

$$E_{av} = \frac{\int_0^{E_F} E \frac{dn}{dE} dE}{\int_0^{E_F} \frac{dn}{dE} dE} = \frac{\int_0^{E_F} E^{\frac{3}{2}} dE}{\int_0^{E_F} E^{\frac{1}{2}} dE} = \frac{3}{5} E_F \approx 20 \text{ MeV} \quad (25)$$

The energy which corresponds to the average momentum is

$$E' = \frac{p_{av}^2}{2m} = \frac{9}{16} \frac{p_F^2}{2m} = \frac{9}{16} E_F \approx 18.6 \text{ MeV} \quad (26)$$

Total energy in a nucleus

Knowing the average energy of a nucleon we can calculate the total energy of a nucleus

$$E_{tot} = N \langle E(p) \rangle + Z \langle E(n) \rangle = N \frac{3}{5} E_F(p) + Z \frac{3}{5} E_F(n)$$

Assuming for simplicity equal mass for a proton and a neutron

$$E_{tot} = \frac{3}{10} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \frac{\hbar^2 c^2}{mc^2 R_0^2} \frac{N^{\frac{5}{3}} + Z^{\frac{5}{3}}}{A^{\frac{2}{3}}}$$

Let us investigate the term which depends on N, Z and A further as it leads to some interesting consequences.

The term dependent on N, Z and A expands to

$$\frac{N^{\frac{5}{3}} + Z^{\frac{5}{3}}}{A^{\frac{2}{3}}} = A \left(1 + \frac{5}{9} \left(\frac{N-Z}{A} \right)^2 \right)$$

Therefore, the total energy has two terms

$$E_{tot} = \frac{3}{10} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \frac{\hbar^2 c^2}{mc^2 R_0^2} A + \frac{1}{6} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} \frac{\hbar^2 c^2}{mc^2 R_0^2} \left(\frac{N-Z}{A} \right)^2$$

The first term corresponds to the volume term in the liquid drop model while the last one corresponds to the symmetry energy term in the liquid drop model.

Therefore, based on the Fermi model we can conclude that the symmetry energy term in the liquid drop model is a quantum mechanical effect related to the way fermion occupy allowed states in the proton/neutron potential well.

1.4 Stable Nuclei

Not all combination of neutrons and protons form stable nuclei. In general, light nuclei ($A < 20$) contain equal numbers of neutrons and protons, while in heavier nuclei the proportion of neutrons becomes progressively greater. This is evident in figure as shown below, which is plot of N versus Z for stable nuclides.

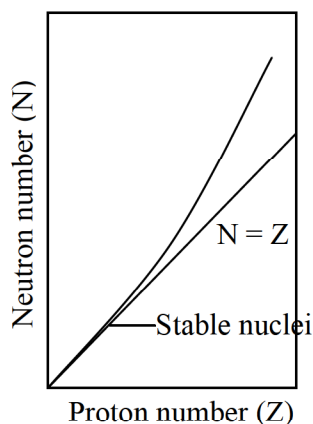


Figure: Neutron-proton diagram for stable nuclei

The tendency for N to equal Z follows from the existence of nuclear energy levels. Nucleons, which have spin $\frac{1}{2}$, obey exclusion principle. Hence proton and neutrons are fermions they follow Pauli exclusion principle. As a result, each energy level can contain two neutrons of opposite spins and two protons of opposite spins. Energy levels in nuclei are filled in sequence, just as energy levels in atoms are, to achieve configurations of minimum energy and therefore maximum stability.

Thus the boron isotope ${}^{12}_5\text{B}$ has more energy than the carbon isotope ${}^{12}_6\text{C}$ because one of its neutrons is in a higher energy level, and ${}^{12}_5\text{B}$ is accordingly unstable. If created in a nuclear reaction, a ${}^{12}_5\text{B}$ nucleus changes by beta decay into a stable ${}^{12}_6\text{C}$ nucleus in a fraction of second.

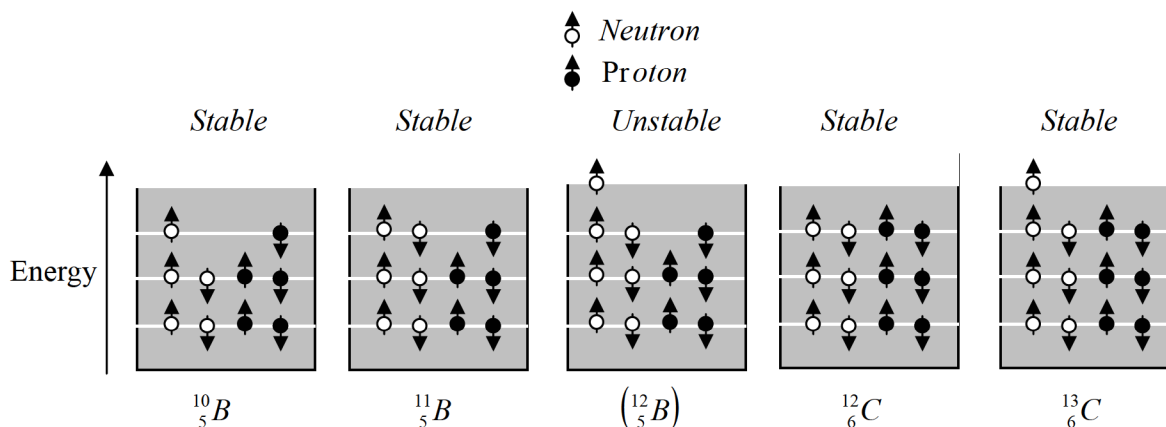


Figure: Simplified energy level diagrams of some boron and carbon isotopes.

The preceding argument is only part of the story. Protons are positively charged and repel one another electrically. This repulsion becomes so great in nuclei with more than 10 protons or so that an excess of neutrons, which produce only attractive forces is required for stability. Thus the curve departs more and more from $N = Z$ line as Z increases.

Sixty percent of stable nuclides have both even Z and N ; these are called “**even-even**” nuclides. Nearly all the others have either even Z and odd N (“**even-odd**” nuclides) or odd Z and even N (“**odd-even**” nuclides) with the numbers of both kinds being about equal.

Only five stable **odd-odd** nuclides are known: ^2_1H , ^6_3Li , $^{10}_5\text{B}$, $^{14}_7\text{N}$ and $^{180}_{73}\text{Ta}$. Nuclear abundances follow a similar pattern of favoring even numbers for Z and N .

These observations are consistent with the presence of nuclear energy levels that can each contain two particles of opposite spin. Nuclei with filled levels have less tendency to pick up other nucleons than those with partially filled levels and hence were less likely to participate in the nuclear reactions involved in the formation of elements.

Nuclear forces are limited in range, and as a result nucleons interact strongly only with their nearest neighbors. This effect is referred to as the **saturation** of nuclear forces. Because the coulomb repulsion of protons is appreciable throughout the entire nucleus, there is a limit to the ability of neutrons to prevent the disruption of large nucleus. This limit is represented by the bismuth isotope $^{209}_{83}\text{Bi}$, which is the **heaviest stable** nuclide.

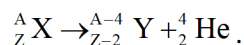
Chapter Three

Radioactivity

Experiments performed subsequently showed that radioactivity was a nuclear phenomenon in which an unstable nucleus undergoes a decay. This is referred to as radioactive decay. Three types of radioactive decay occur in nature:

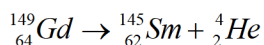
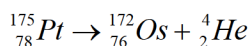
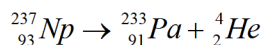
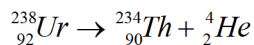
(i) Alpha Decay

α -decay in which a helium nucleus ${}^4_2\text{He}$ is emitted;



Where ${}^A_Z\text{X}$ parents' nucleus, ${}^{A-4}_{Z-2}\text{Y}$ is the daughter and ${}^4_2\text{He}$

A is total no of Nucleons, Z is the total no of proton.



(ii) Beta decay

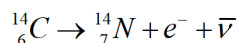
In beta decay, a nucleus spontaneously emits an electron (β^- decay) or a positron (β^+ decay).

(1) Beta-minus decay

In beta minus, a neutron is transformed into proton causing increase of atomic number. The change in atomic configuration is given by ${}_Z^AX \rightarrow {}_{Z+1}^AY + e^- + \bar{\nu}$

To maintain the conservation of charge the nucleus process also an electron and antineutrino
 $N = P + e^- + \bar{\nu}$

A common example of β^- decay is ${}_{15}^{32}P \rightarrow {}_{16}^{32}S + e^- + \bar{\nu}$

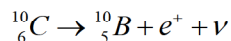


(2) Beta-plus decay

In beta plus, a proton is transformed into neutron causing decrease of atomic number. The change in atomic configuration is given by ${}_Z^AX \rightarrow {}_{Z-1}^AY + e^+ + \nu$

To maintain the conservation of charge the nucleus process also an electron and antineutrino.
 $P = N + e^+ + \nu$

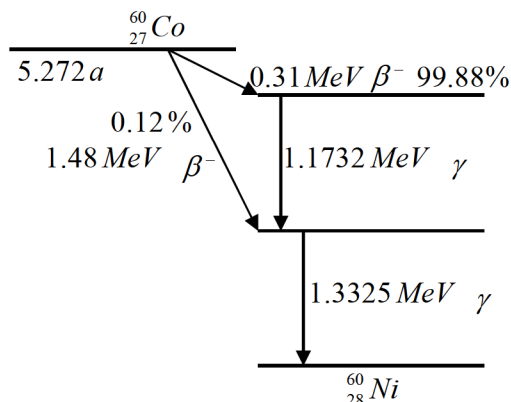
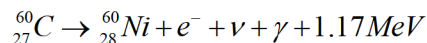
A common example of β^+ decay is ${}_{11}^{22}Na \rightarrow {}_{10}^{22}Ne + e^+ + \nu$

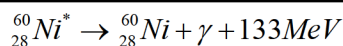


(iii) Gamma Decay

Like an atom, a nucleus also has discrete energy levels - the ground state and excited states. The scale of energy is, however, very different. Atomic energy level spacings are of the order of eV while the difference in nuclear energy levels is of the order of MeV . When a nucleus in an excited state spontaneously decays to its ground state (or to a lower energy state), a photon is emitted with energy equal to the difference in the two energy levels of the nucleus. This is the so-called gamma decay. The energy MeV corresponds to radiation of extremely short wavelength shorter than X ray region. Typically, a gamma ray is emitted when a α or β decay results in a daughter nucleus in an excited state. This then returns to the ground state by a single photon transition or successive transition.

Example of gamma decay





When a beam of γ rays photons passes through matter, the intensity of beam decreases exponentially $I = I_0 \exp(-\mu x)$ where I_0 : Initial Intensity, μ : absorption coefficient of substance, x : thickness of Matter.

Radioactivity

A. H. Becquerel discovered radioactivity in 1896 purely by accident. While studying the fluorescence and phosphorescence of compounds irradiated with visible light, Becquerel observed an interesting phenomenon. After illuminating some pieces of uranium-potassium sulphate with visible light, he wrapped them in black paper and separated the package from a photographic plate by a piece of silver. When, after several hours of exposure, the photographic plate was developed, it showed blackening due to something that must have been emitted by the compound and was able to penetrate both black paper and the silver.

Law of radioactivity

If N is number of nuclei in the sample and ΔN number of nuclei decay in time Δt then

$$\frac{\Delta N}{\Delta t} \propto N \Rightarrow \frac{\Delta N}{\Delta t} = -\lambda N \text{ where } \lambda \text{ is proportionality constant known as decay constant or}$$

disintegration constant, the negative sign arises because number of nuclei in sample will decrease as the time increase. dN is the change in N , which may have either sign. Here it is negative, because out of original N nuclei, ΔN have decayed, leaving $(N - \Delta N)$ nuclei.

$$\frac{\Delta N}{\Delta t} = -\lambda N \text{ as } \Delta t \rightarrow 0 \frac{dN}{dt} = -\lambda N \text{ which is first order differential equation.}$$

$$\int_{N_0}^N \frac{dN}{N} = - \int_{t_0}^t dt \Rightarrow \ln N - \ln N_0 = -(t - t_0) \text{ set } t_0 = 0 \Rightarrow N = N_0 \exp(-\lambda t)$$

Decay Rate (Activity)

The term $R = -\frac{dN}{dt}$ is defined as rate of decay.

$$\frac{dN}{dt} = -\lambda N \Rightarrow -\frac{dN}{dt} = R = \lambda N = \lambda N_0 \exp(-\lambda t)$$

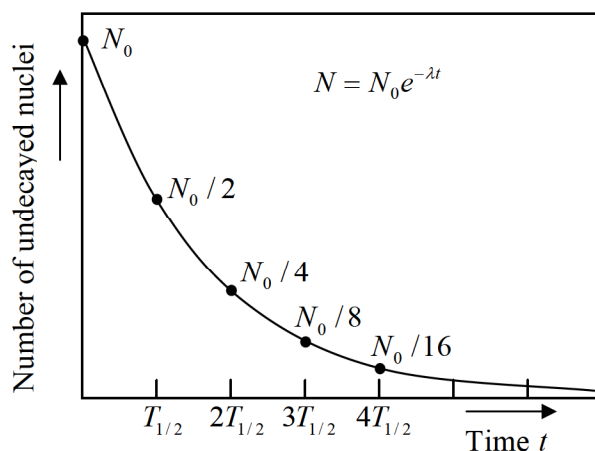


Figure: Exponential decay of a radioactive species. After a lapse of $T_{1/2}$, population of the given species drops by a factor of 2.

where $\lambda N_0 = R_0$ is initial rate or decay rate at $t = 0$ so $R = R_0 \exp(-\lambda t)$

The decay rate of a sample, rather than the number of radioactive nuclei, is a more direct experimentally measurable quantity and is given a specific name: activity. The SI unit for activity is becquerel, named after the discoverer of radioactivity, Henry Becquerel.

Unit of Activity or decay rate $1 \text{ curie} = 1 \text{ ci} = 3.7 \times 10^{10} \text{ decay per sec} = 3.7 \times 10^{10} \text{ Bq}$

Half-life: Half-life of Radionuclide denoted by $t_{1/2}$ is the time it takes for sample that has initially say N_0 radio nuclei to reduce to $\frac{N_0}{2}$ so

$$N = N_0 \exp(-\lambda t) \Rightarrow \frac{N_0}{2} = N_0 \exp(-\lambda t_{1/2}) \Rightarrow t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Clearly if N_0 reduces to half value in time $t_{1/2}$ R_0 will also reduce to $\frac{R_0}{2}$ in time $t_{1/2} = \frac{0.693}{\lambda}$.

Average Mean life: The average life time of radioactive particle before it decay. although they are constant, they are associated with statistical behaviour of population of atom. the life time is given by

$$\tau = \frac{\text{total lifetime of nuclei}}{\text{Total number of nuclei}} = \frac{\int_0^\infty t dN}{N_0} = \frac{\int_0^\infty t \lambda N_0 \exp(-\lambda t) dt}{N_0} = \lambda \int_0^\infty t \exp(-\lambda t) dt \Rightarrow \tau = \frac{1}{\lambda}$$

So $t_{1/2} = 0.693 \cdot \tau$

Radioactivity Equilibrium: If we start with a sample of N_A nuclei with parent nuclide A , after a period of time an equilibrium situation will come about in which each successive daughter $B, C, \dots$ decays at the same rate as it is formed. Thus the activities $R_A, R_B, R_C, \dots$ are equal at equilibrium since $R = \lambda N$ we have radioactive equilibrium $N_A \lambda_A = N_B \lambda_B = N_C \lambda_C = \dots$ where each N_A, N_B and N_C decrease exponentially with time.

Example: The half life of ${}^{238}_{92}\text{U}$ undergoing α decay is 4.5×10^9 years. Find the activity of 1g of sample of ${}^{238}_{92}\text{U}$.

Solution: $t_{1/2} = 4.5 \times 10^9 \text{ Y} = 4 \times 10^9 \times 3.16 \times 10^7 \text{ sec/Y} = 1.42 \times 10^{17} \text{ sec}$

1 gram of uranium consists of $\frac{1}{238 \times 10^{-3}} \text{ kmol} \times 6.025 \times 10^{26} \text{ atom / kmol}$

$$1 \quad R = \lambda N = \frac{0.693}{t_{1/2}} \cdot N = \frac{0.693}{1.42 \times 10^{17}} \times 25 \times 10^{20} \text{ sec}^{-1}$$

$$1.23 \times 10^4 \text{ sec}^{-1} = 1.23 \times 10^4 \text{ Bq}$$

Example: Tritium has half life of 12.5 Y undergoes beta decay. What fraction of sample of pure tritium will remain undecayed after 25 Y

Solution: Let us assume initial atom is N_0 in first 12.5 there will be $\frac{N_0}{2}$ atom will decay so

$$N_0 - \frac{N_0}{2} = \frac{N_0}{2} \text{ will remain.}$$

Now in next 12.5 year half of atom of $\frac{N_0}{2} = \frac{N_0}{4}$ will remain undecayed

Example: The half life of strontium ${}^{90}_{38}\text{Sr}$ is 28.8 Y

(a) Find decay constant

(b) Find initial activity of 1.00g of material

(c) Find the activity after 59.6 Y

Solution: (a) Decay constant is $\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{28.8 \times 3.16 \times 10^7 \text{ sec}} = 7.61 \times 10^{-10} \text{ sec}^{-1}$

(b) The atomic mass of ${}^{90}_{38}\text{Sr}$ is 89.91 g Avogadro number is given by $6.022 \times 10^{23} \text{ atom / mol}$

number of atom in 1g of ${}^{90}_{38}\text{Sr}$ is $\frac{1.00 \text{ g}}{89.91 \text{ g}} \times 6.02 \times 10^{23} \text{ nuclei}$

initial activity, $R_0 = \lambda N_0 = (7.61 \times 10^{-10} \text{ sec}^{-1}) \times (6.70 \times 10^{21} \text{ nuclei}) = 5.10 \times 10^{12} \text{ decay/sec}$

$$(c) R = \lambda N(t) = \lambda \times N_0 \times \exp(-\lambda t) = \lambda N_0 \exp\left(-\left(0.693 \frac{t}{t_{1/2}}\right)\right)$$

$$t = 59.6Y \quad t_{1/2} = 28.8Y$$

$$R(t) = \lambda N_0 \exp\left(-\left(0.693 \frac{t}{t_{1/2}}\right)\right) = R_0 \exp(-0.693 \times 2) = R_0 \times \exp(-1.386) = 0.25 R_0$$

$$= 0.25 \times 5.10 \times 10^{12} \text{ decay/sec} = 1.275 \text{ decay/sec}$$

Example: Approximately 20% of human body by mass is carbon. Calculate the activity due to $^{14}_6\text{C}$ in 1 kg of carbon found in living organism. Express the activity in the unit of *Bq* and *Ci*.

Solution: The abundance of ^{14}C in carbon sample is 1.3×10^{-12} . half-life of ^{14}C is 5730 year.

Number of carbon ^{12}C nuclei in 1kg of sample

$$N(^{12}\text{C}) = \frac{6.02 \times 10^{23} \text{ mol}^{-1}}{12.0 \text{ g/mol}} \times 1000 \text{ g} = 5.02 \times 10^{25}$$

$$\text{Number of carbon } N(^{14}\text{C}) = (5.02 \times 10^{25}) (1.3 \times 10^{-12}) = 6.52 \times 10^{12}$$

Now we can find the activity *A* by using the equation

$$A = \lambda N = \frac{0.693}{t_{1/2}} N = \frac{0.693}{5730} \times 6.52 \times 10^{12} \text{ y}^{-1} = \frac{7.89 \times 10^9 \text{ y}}{3.16 \times 10^7 \text{ s}} = 250 \text{ Bq}$$

$$A = \frac{250}{3.7 \times 10^{10} \text{ Bq/Ci}} = 6.76 \times 10^{-9} \text{ Ci} \Rightarrow 6.76 \text{ nCi}$$

Example: In an ancient burial cave, a team of archaeologist discovers ancient wood furniture.

Only 80% of the original ^{14}C remains in wood. How old is furniture? Half life of ^{14}C is 5730 year.

Solution: $\frac{N}{N_0} = \exp(-\lambda t)$ it is given $\frac{N}{N_0} = 0.80 \Rightarrow \exp(-\lambda t) \Rightarrow -\lambda t = \ln 0.80 = -0.223 = -\lambda t$

$$\text{So } t = \frac{0.223}{\lambda} \Rightarrow \frac{0.223}{\frac{0.693}{t_{1/2}}} = \frac{0.223}{0.693} \times 5730 \text{ y} = 1844 \text{ y}$$

Example: The atomic ratio between the uranium isotopes ^{238}U and ^{234}U in a mineral sample is found to be 1.8×10^4 . The half-life of ^{234}U is 2.5×10^5 y. Find the half-life of ^{238}U .

Solution: $N(238) \times \lambda(238) = N(234) \lambda(234) \Rightarrow N(238) \times \frac{0.693}{t_{1/2}(238)} = N(234) \times \frac{0.693}{t_{1/2}(234)}$

$$= t_{1/2}(238) = \frac{N(238)}{N(234)} \times t_{1/2}(234) \Rightarrow t_{1/2}(238) = (1.8 \times 10^4)(2.5 \times 10^5 \text{ y}) = 4.5 \times 10^9 \text{ y}$$

Example: A substance X disintegrates into substance Y find the number of nuclei of Y which is expected number of nuclei N_Y still undecayed. Where N_{X_0} and N_{Y_0} are initial amount of X and Y respectively. λ_x and λ_y are decay constant for X and Y .

Solution: $\frac{dN_x}{dt} = -\lambda_x N_x \Rightarrow \frac{dN_x}{dt} = -\lambda_x N_{X_0} \exp(-\lambda_x t)$

For X nucleus disintegrate with rate of $\lambda_x N_x$ which is also rate at which Y is formed .so decay

equation for Y is given by $\frac{dN_Y}{dt} = \lambda_x N_x - \lambda_y N_Y \Rightarrow \frac{dN_Y}{dt} = \lambda_x N_{X_0} \exp(-\lambda_x t) - \lambda_y N_Y$

$$N_Y \lambda_y + \frac{dN_Y}{dt} = N_{X_0} \lambda_x \exp(-\lambda_x t) \Rightarrow N_Y \lambda_y dt + dN_Y = N_{X_0} \lambda_x \exp(-\lambda_x t) dt$$

Multiply both side by $\exp(\lambda_y t)$

$$\exp(\lambda_y t) N_Y \lambda_y dt + \exp(\lambda_y t) dN_Y = N_{X_0} \lambda_x (\exp \lambda_y t) \exp(-\lambda_x t) dt$$

$$d(N_Y \exp(\lambda_y t)) = N_{X_0} \lambda_x (\exp \lambda_y t) \exp(-\lambda_x t) dt$$

$$N_Y \exp(\lambda_y t) = \frac{N_{X_0} \lambda_x}{\lambda_y - \lambda_x} \exp(\lambda_y t - \lambda_x t) + C \text{ at } t = 0, N_Y = N_{Y_0} \Rightarrow C = N_{Y_0} - \frac{\lambda_x N_{X_0}}{\lambda_y - \lambda_x}$$

$$N_Y = \left(N_{Y_0} - \frac{\lambda_x N_{X_0}}{\lambda_y - \lambda_x} \right) \exp(-\lambda_y t) + \left(\frac{\lambda_x N_{X_0}}{\lambda_y - \lambda_x} \right) \exp(-\lambda_x t)$$

Nuclear Reaction: Nuclear reaction is a process in which two nuclei, or a nucleus and an external subatomic particle, collide to produce one or more new nuclides. Thus, a nuclear reaction must cause a transformation of at least one nuclide to another. If a nucleus interacts with another nucleus or particle and they then separate without changing the nature of any nuclide, the process is simply referred to as a type of nuclear scattering, rather than a nuclear reaction.

Typically, nuclear reaction represented in term of equation as $x + X = Y + y$ where x projectile hits the target X and Y and y are out going particle.

The Q value is defined as total change in kinetic energy.

Total energy of particle X is given by $E_X = T_X + M_X c^2$ where T_X is kinetic energy and $M_X c^2$ is rest mass energy

Total energy of particle x is given by $E_x = T_x + M_x c^2$ where T_x is kinetic energy and $M_x c^2$ is rest mass energy

Total energy of particle Y is given by $E_Y = T_Y + M_Y c^2$ where T_Y is kinetic energy and $M_Y c^2$ is rest mass energy

Total energy of particle y is given by $E_y = T_y + M_y c^2$ where T_y is kinetic energy and $M_y c^2$ is rest mass energy

From the conservation energy $E_X + E_x = E_Y + E_y$

$$(T_X + M_X c^2) + (T_x + M_x c^2) = (T_Y + M_Y c^2) + (T_y + M_y c^2)$$

The Q value is given by

$$(T_Y + T_y) - (T_X + T_x) = [(M_X + m_x) - (M_Y + M_y)] c^2$$

So
$$Q = [(M_X + m_x) - (M_Y + M_y)] c^2$$

This equation will mainly valid for one dimensional collision.

Some other facts about Q value

(1) The KE_α of the emitted α -particle is never quite equal to Q , since momentum must be conserved, the nucleus recoils with a small amount of kinetic energy when the α -particle emerges. Thus $KE_\alpha \approx \frac{A-4}{A} \cdot Q$

(2) Center of mass coordinate system

Most of nuclear reaction in a laboratory occur when a moving nucleons or nucleus strikes a stationary one. analysing such a reaction is simplified when we use a coordinate system that moves with the center of mass of colliding particles.

The observer located at center of mass, the particle has equal and opposite momenta .Hence particle of mass m_A with speed v approaches to stationary particle of mass m_B . As viewed from laboratory ,the speed v_{cm} is speed of center of mass

$$m_A(v - v_{cm}) = m_B v_{cm} \Rightarrow v_{cm} = \left(\frac{m_A}{m_A + m_B} \right) v$$

In most of the nuclear reaction takes place $v \ll c$ so nonrelativistic treatment is sufficient

Kinetic energy in lab frame is given by $T_{lab} = \frac{1}{2} m_A v^2$

Kinetic energy in center mass frame is given by $T_{cm} = \frac{1}{2} m_A (v - v_{cm})^2 + \frac{1}{2} m_B v_{cm}^2$

$$= \frac{1}{2} m_A v^2 + \frac{1}{2} m_B v_{cm}^2 - \frac{1}{2} 2 m_A v v_{cm} + \frac{1}{2} m_B v_{cm}^2$$

$$= \frac{1}{2} m_A v^2 + \frac{1}{2} m_B v_{cm}^2 - \frac{1}{2} 2 m_A \left(\frac{m_A + m_B}{m_A} \right) v_{cm} v_{cm} + \frac{1}{2} m_B v_{cm}^2$$

$$\frac{1}{2} m_A v^2 - \frac{1}{2} (m_A + m_B) v_{cm}^2 = \frac{1}{2} m_A v^2 - \frac{1}{2} (m_A + m_B) \left(\frac{m_A}{m_A + m_B} \right)^2 v^2$$

$$= \frac{1}{2} m_A v^2 - \frac{1}{2} \left(\frac{m_A}{m_A + m_B} \right) m_A v^2$$

$$T_{cm} = T_{lab} - \frac{m_A}{m_A + m_B} T_{lab} \Rightarrow T_{cm} = \left(\frac{m_B}{m_A + m_B} \right) T_{lab}$$

Here we can see that $T_{cm} < T_{lab}$

The Q value of nuclear reaction $A + B \rightarrow C + D$ is given by $Q = (m_A + m_B - m_C - m_D) c^2$

If $Q > 0$ K.E. of product is **more than** kinetic energy of reactant, which means the reaction is exothermic reaction. The energy is given off by the reaction.

If $Q < 0$ K.E. of product is **less than** kinetic energy of reactant, which means the reaction is endothermic reaction. The energy will flow into system.

Which means enough kinetic energy KE_{cm} in the center of mass frame must be provided by the reacting particle so that $KE_{cm} + Q \geq 0$

Example: The polonium isotope $^{210}_{84}\text{Po}$ is unstable and emits a 5.30MeV alpha particle. The atomic mass of $^{210}_{84}\text{Po}$ is $209.989u$ and that of ^4_2He is $4.006u$. Identify the daughter nuclei and find its mass.

Solution: $^{210}_{84}\text{Po} = ^{206}_{82}\text{Po} + ^4_2\text{He}$ so new particle is $^{206}_{82}\text{Po}$ let us assume its mass is m_f .

The kinetic energy of alpha particle is $KE = 5.34\text{MeV}$

So, we can calculate the Q Value. $Q = \frac{A}{A-4} K.E_\alpha = \left(\frac{210}{210-4} \right) 5.30 \text{ MeV} = 5.40 \text{ MeV}$

The equivalent mass is given by $\frac{5.40 \text{ MeV}}{931 \text{ MeV/u}} = 0.0058 \text{ u}$

So $m_Q = m_i - m_\alpha - m_f \Rightarrow m_f = m_i - m_\alpha - m_Q$

$$209.9829 \text{ u} - 4.0026 \text{ u} - 0.0058 \text{ u} = 205.9745 \text{ u}$$

Example: Find the minimum kinetic energy in laboratory frame needed by an alpha particle to cause the reaction $^{14}\text{N}(\alpha, p)^{17}\text{O}$. The masses of ^{14}N , ^4He , ^1H and ^{17}O are respectively 14.00307 u , 4.00260 u , 1.00783 u and 16.99913 u

Solution: The reaction is $^{14}\text{N}(\alpha, p)^{17}\text{O}$ which means $^4_2\text{He} + ^{14}_7\text{N} \rightarrow ^{17}_8\text{O} + p$

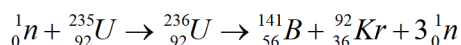
$Q = (14.00307 \text{ u} + 4.00260 \text{ u} - 1.00783 \text{ u} - 16.99913 \text{ u}) \times 931 \text{ MeV/u} = -1.20 \text{ MeV}$ which is negative so reaction is endothermic in nature.

The minimum kinetic energy in centre of mass frame is $T_{cm} = 1.20 \text{ MeV}$ in order to reaction

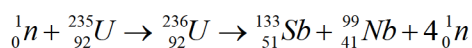
occur. $T_{lab} = \left(\frac{m_A + m_B}{m_B} \right) T_{cm} = \left(\frac{4.00260 + 14.00307}{14.00307} \right) 1.20 \text{ MeV} = 1.54 \text{ MeV}$

Nuclear fission: Nuclear fission is a nuclear reaction in which the nucleus of an atom breaks into two or more smaller nuclei. The fission process often produces gamma photon, and releases a very large amount of energy even by the energetic standards of radioactive.

Some important nuclear fission



The same reaction can produce other pair of intermediate mass fragments



Another example $^1_0n + ^{235}_{92}\text{U} \rightarrow ^{140}_{54}\text{Xe} + ^{94}_{38}\text{Sr} + 2^1_0n$

Example: For the reaction $^1_0n + ^{235}_{92}\text{U} \rightarrow ^{236}_{92}\text{U} \rightarrow ^{141}_{56}\text{Ba} + ^{92}_{36}\text{Kr} + 3^1_0n$, find how much energy released in the process.

Solution: Mass of $^{235}_{92}\text{U} = 235.045733 \text{ u}$, Mass of $^1_0n = 1.008665 \text{ u}$

Total initial mass = 236.054398 u .

Mass of $^{141}_{56}\text{Ba} = 140.99177 \text{ u}$, Mass of $^{92}_{36}\text{Kr} = 91.8854 \text{ u}$, Mass of $3^1_0n = 3.0259 \text{ u}$,

Total final mass = 235.00307 u .

Mass defect (Δm) = initial mass - final mass = 1.051328 u .

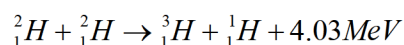
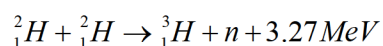
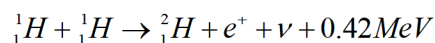
Energy released i.e. Q -value = $\Delta m \times 979.312032 \text{ MeV} \approx 200 \text{ MeV}$.

Number of uranium nucleus in 1 kg of uranium = $6.023 \times 10^{23} \times 1000/235$.

Energy released by 1 kg of uranium = $6.023 \times 10^{23} \times 1000/235 \times 200 \text{ MeV} = 5.13 \times 10^{26} \text{ MeV}$

Nuclear fusion is a nuclear reaction in which two or more atomic nuclei are combined to form one or more different atomic nuclei and subatomic particles (neutrons or protons). The difference in mass between the reactants and products is manifested as either the release or the absorption of energy.

Some important nuclear fusion reaction



Nuclear Fusion in Stars

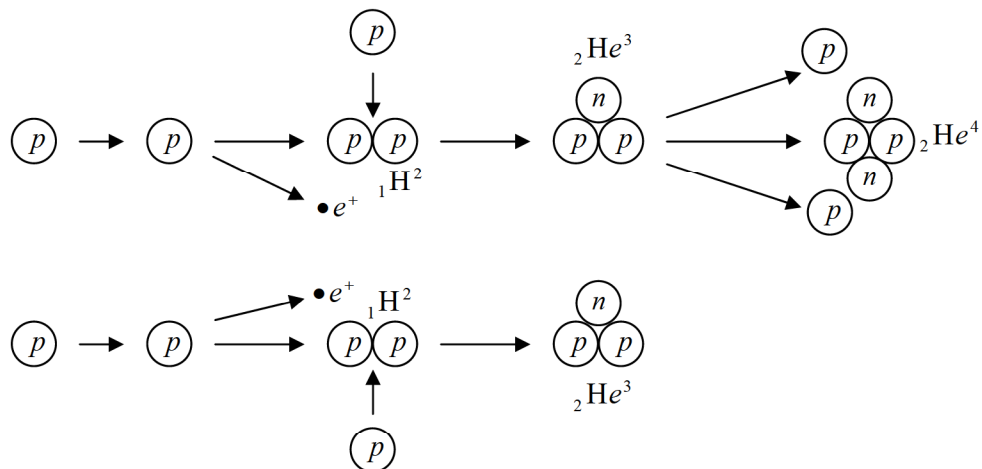
Fusion is a process, in which the lighter nuclei fuse together and produce a heavier nucleus. The sum of the masses of the individual light nuclei is more than would be the mass of the nucleus formed by their fusion, and thus the fusion process result in liberation of energy.

Self-sustaining fusion reactions can occur only under conditions of **extreme temperature** and **density**. The high temperature ensures that the same nuclei have the energy needed to come close enough together to interact, and the high density ensures that such collisions are frequent. A further condition for the proton-proton cycles is a large reacting mass, such as that of the sun, since much time may elapse between the initial fusion of a particular proton and its eventual incorporation in an α -particle.

The basic energy producing process in the sun is the fusion of hydrogen nuclei into helium nuclei. This can take place in several different reaction sequences, the most common of which, the **proton–proton cycle**. The total evolved energy is 24.7 MeV per ${}_2^4\text{He}$ nucleus formed.

Since 24.7 MeV is $4 \times 10^{-12} \text{ J}$, the sun's power output of $4 \times 10^{26} \text{ W}$ means each sequence of reactions must occur 10^{38} times per second.

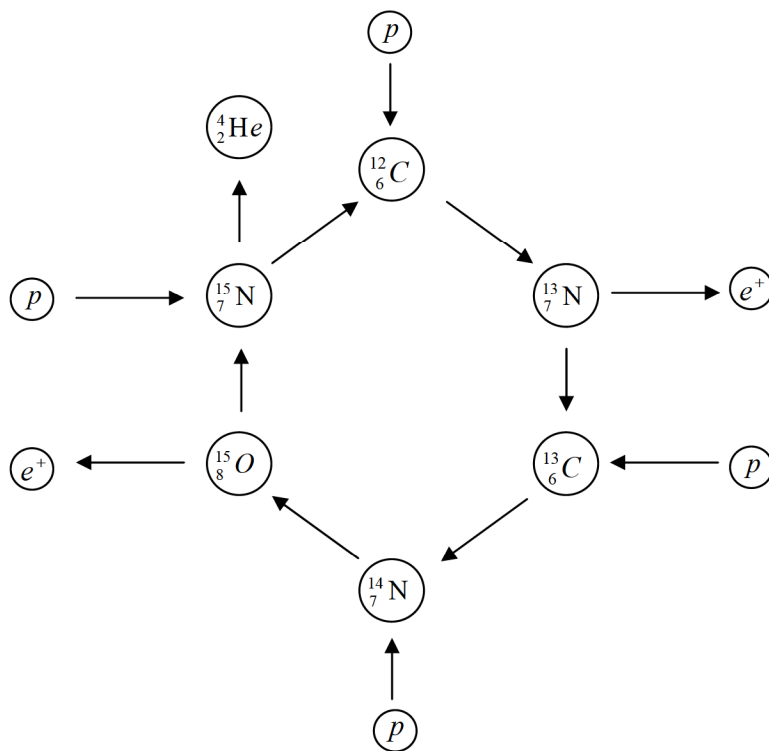
- Proton-Proton cycle**



Energy is given off at each step. The net result is the combination of four hydrogen nuclei to form a helium nucleus and two positrons.

- Carbon-Cycle**

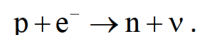
It also involves the combination of four hydrogen nuclei (Stars hotter than sun) to form a helium nucleus with the evolution of energy. The net result again is the formation of an α -particle and two positrons from four protons, with the evolution of 24.7 MeV. The initial ${}^{12}_6\text{C}$ acts as a kind of catalyst for the process, since it reappears at its end.



Some other nuclear reaction

Electron capture

It is closely connected with positron emission. In electron capture a nucleus absorbs one of its inner atomic electrons, with the result that a nuclear proton becomes a neutron and neutrino is emitted:



Usually, the absorbed electrons come from the *K*-shell, and an X-ray photon is emitted when one of the atoms outer electrons falls into the resulting vacant state. The wavelength of the photon will be one of those characteristics of daughter element, not of the original one, and the process can be recognized on that basis.

Note: 1. Electron capture is competitive with positron emission since both processes lead to the same nuclear transformation.

2. Electron capture occurs more often than positron emission in heavy nuclides because the electrons in such nuclides are relatively close to the nucleus, which promotes their interaction with it.

Inverse Beta Decay: In this reaction proton will absorb antineutrino and produce neutron and positron or neutron will absorb neutrino and produce electron. they can be shown as reaction respectively $p + \bar{\nu} = n + e^+$ and $n + \nu = p + e^-$

These reactions are very low probability which is why neutrinos and antineutrino are able to pass vast amount of matter.